

## Quiz 2: Qualitative Analysis I — The Workflow

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*Review of Session 2. ~10–15 minutes.*

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### Section A: The battery

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1. List the eight powder tests in the order you run them.
2. Why does running the tests in a fixed order make you faster?
3. How much sample should you use per test?

### Section B: Reading results

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1. Two powders both fizz with acid. One dissolves fully, one stays cloudy. Name each (with formula).
2. A powder makes an "oobleck" — liquid when stirred slowly, solid when poked fast. What is it, and what will iodine do to it?
3. A powder forms cubic crystals, dissolves fully, doesn't fizz, no iodine reaction, and crackles without melting on gentle heat. What is it?
4. What are the three parts of an ID answer that earn full points?

### Section C: Dichotomous keys

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1. In a key for {baking soda, chalk, cornstarch, salt, sugar, gypsum}, which single test separates the most powders at once?
2. Trace a key: a powder stays cloudy in water, does not fizz with acid, and turns iodine blue-black. What is it?

### Section D: Mixtures

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1. Name two signs that a "sample" is actually a mixture of two powders.
2. If a suspected mixture is really one powder, how do you know?

## Answer Key

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### Section A

1. Look, smell, feel, solubility in water, pH, reaction with acid, reaction with iodine, heat.
2. No decisions between tests; you can set up all samples and batch one test across all of them; you don't forget a test.
3. A pinch — about what sticks to a wet toothpick.

### Section B

1. Dissolves fully = baking soda,  $\text{NaHCO}_3$ . Stays cloudy = calcium carbonate (chalk),  $\text{CaCO}_3$ .
2. Cornstarch,  $(\text{C}_6\text{H}_{10}\text{O}_5)_n$ ; iodine turns it blue-black (starch).
3. Table salt ( $\text{NaCl}$ ).
4. Name, chemical formula, and the suspect it implicates (with the reason).

### Section C

1. The acid-fizz test — it isolates the two carbonates immediately.
2. Cornstarch (not a carbonate → not step 2; iodine positive → cornstarch).

### Section D

1. Two crystal shapes or colors under the lens; a partial test result (some fizz *and* residue; iodine blackens part and leaves grains orange; half-dissolves).
2. There is a single powder that explains *all* of the observations. If no single powder explains everything, it's a mixture.